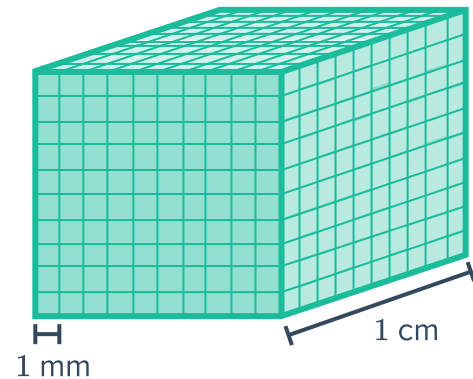


Units of volume and capacity



- 1 The cube shown in the diagram has a volume of 1 cm^3 . Find its volume in cubic millimetres.



- 2 Convert the following to cubic millimetres:

a 3.5 cm^3 b 4.2 m^3 c 4 m^3 d 19 cm^3

- 3 Convert the following to cubic centimetres:

a $47\,600 \text{ mm}^3$ b 8.12 m^3 c 7000 mm^3 d 33 m^3
e 1 mL f 8000 mL g 40 mL h 4.47 mL

- 4 What operation is required to convert cubic centimetres to cubic metres?

- 5 Convert the following to cubic metres:

a $1\,070\,000\,000 \text{ mm}^3$ b $51\,980\,000 \text{ cm}^3$ c $460\,000\,000 \text{ mm}^3$ d $16\,000\,000 \text{ cm}^3$
e 6000 L f 800 L g 18 kL h 0.95 kL

- 6 Which metric unit represents 'one thousandth of a litre'?

- 7 Which metric unit represents 'one thousand millilitres'?

- 8 Convert the following to millilitres:

a 6 L b 1.54 L c 2 kL d 0.17 kL
e 1 cm^3 f 8.5 cm^3 g 9000 mm^3 h 0.005 m^3

9 Convert the following to litres:



- | | | | |
|------------------------|--------------------|------------------------|-----------------------|
| a 4920 mL | b 5000 mL | c 9 kL | d 1.67 kL |
| e 18.62 m ³ | f 2 m ³ | g 6750 cm ³ | h 800 cm ³ |

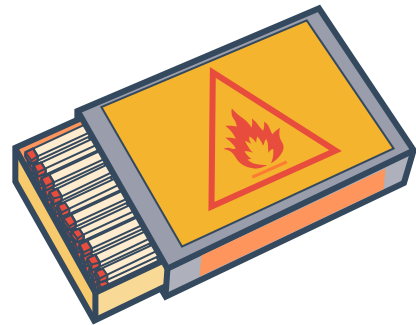
10 Convert the following to kilolitres:

- | | | | |
|---------------------|-----------------------|------------------------------|--------------------------|
| a 6570 L | b 84.6 L | c 1 500 000 mL | d 60 000 mL |
| e 17 m ³ | f 0.37 m ³ | g 25 000 000 cm ³ | h 54 000 cm ³ |

Volume estimation

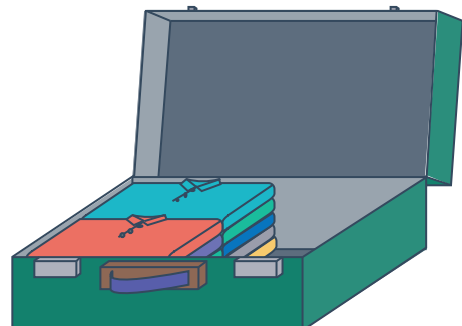
11 Consider the matchbox shown:

- a State the most appropriate unit for measuring the volume of a match box.
- b Estimate the volume of the matchbox.



12 The volume of the suitcase shown has a total capacity of 88 500 cm³.

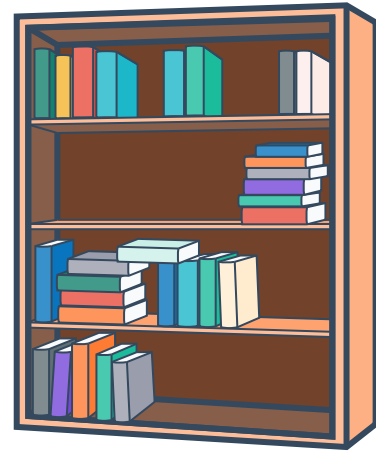
Estimate the volume of the contained shirts.



- 13 The volume of the bookshelf shown below is known to be 0.64 m^3 .

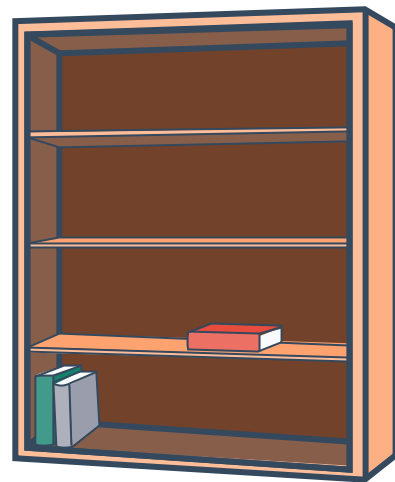


Estimate the volume of the books on the bookshelf.



- 14 The volume of the bookshelf shown is known to be 0.68 m^3 .

Estimate the volume of the books on the bookshelf.



- 15 State the most appropriate unit for measuring the volume of the following:

- | | | | |
|-----------------|----------------|--------------------|-----------------|
| a Sand pit | b Wardrobe | c Water glass | d Swimming pool |
| e Air in a hall | f Water bottle | g Air in a balloon | |

Capacity estimation

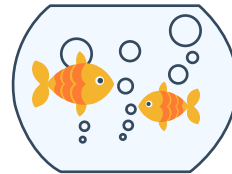
- 16 Which units can we measure with a jug like the one in the picture?



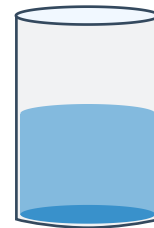
- 17 State the most appropriate unit for measuring the capacity of a cup of coffee.



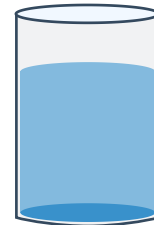
- 18 State the most appropriate unit for measuring the capacity of a fish bowl.



- 19 Estimate the volume of liquid in this container if it is able to hold 2400 mL when full.



- 20 Estimate the volume of liquid in this container if it is able to hold 3.6 L when full.



- 21 Estimate the volume of liquid in this container if it is able to hold 3600 mL when full.



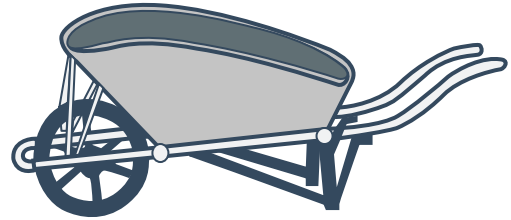
- 22 Estimate the volume of liquid in this container if it is able to hold 3.8 litres when full.





23 A landscaper needs to move soil across the garden. Choose the best estimate for the capacity of soil that can be carried by a wheelbarrow in one trip.

- A** 771 mL **B** 21 ML **C** 71 L



24 State the most appropriate unit of measure for finding the volume of the following:

- | | |
|----------------------------------|---|
| a Water in a dam. | b A dose of vaccine injection. |
| c A small carton of milk. | d The Atlantic Ocean. |
| e A dosage of medicine. | f A bottle of washing detergent. |

Applications

25 Quiana bought a 400 mL bottle of soft drink. How many litres of soft drink did she buy?

26 The following baby bottle holds 100 mL of milk:

Find its volume in cm^3 .



27 The following water bottle has a volume of 700 cm^3 :

- a** Find its capacity in millilitres.
b Find its capacity in litres.



28 This juice box contains 1600 mL of juice:



- a** Find its volume in cm^3 .
- b** Calculate the length of the juice box if it is 8 cm tall and 5 cm wide.



29 A cubic box has capacity of 343 L. Find its volume in cubic centimetres.

30 A small pond contains 3900 L of water. Find the capacity of the pond in cubic metres.

31 A jug has a volume of $12\,395\text{ cm}^3$. How many litres of soda water can it hold? Round your answer to three decimal places.